Cook Along: testing phase plan

Goal: find and fix the bugs behind "the timer isn't on the pan" using automated tests, before real footage is needed. Scope is everything built so far, with the AR anchoring pipeline first.

How to run this

- **Run one sub-phase per session.** Start a fresh session (or `/clear`) and say:

```
Run sub-phase 1.1 from docs/testing-plan.md
```

A fresh session reads only this file and the code it needs, which costs far less than continuing one long conversation.

- **Every sub-phase appends its results to `docs/testing-log.md`.** The next sub-phase reads the log, so nothing needs re-explaining.
- **Set the model and effort before starting** (model picker in the app, or `/model`). Recommendations are in the table.
- **No deploys in any sub-phase.** Commits are local only, and only where noted.
- **Kitchen photos stay out of git.** Test fixtures built from your pan photos go in `tests/fixtures/private/`, which is gitignored.

Model and effort key

Model / effort	Use it for
Opus 5.5, high	subtle logic, coordinate maths, deciding root causes
Opus 5.5, medium	focused fixes once the cause is known
Sonnet 5.5, medium	writing test code, running tests, reading results
Sonnet 5.5, low	small mechanical changes, summaries
Haiku 5.5, low	installs, commits, file shuffling

Plan

Priority: **P1** = on the critical path to fixing anchoring. **P2** = valuable, skippable if short on time. **P3** = nice to have.

ID	Sub-phase	What it's for	Est.	Model	Effort	Pri
0.1	Snapshot	Commit current state to a `testing-phase` branch so every fix is reviewable and reversible	5m	Haiku 5.5	low	P1
1.1	Code review: AR pipeline	Find bugs in detection, tracking, mapping and rendering before writing any tests	30m	Opus 5.5	high	P1
1.2	Dead code inventory	List leftovers from the four earlier tracker attempts, safe to delete or not	10m	Sonnet 5.5	low	P3

ID	Sub-phase	What it's for	Est.	Model	Effort	Pri
2.1	Install Playwright	Chromium and WebKit test browsers	10m	Haiku 5.5	low	P1
2.2	Synthetic test clips	Animate your pan photos with known motion, so the pan's true position is known in every frame	25m	Sonnet 5.5	medium	P1
3.1	Playwright config	iPhone 16 Pro viewport, touch, fake camera fed from the clips	20m	Sonnet 5.5	medium	P1
3.2	Mock the vision route	Scripted Gemini answers: accurate, wrong, empty, failing, slow. Reproducible, uses no quota	20m	Sonnet 5.5	medium	P1
3.3	Measurement	Per-frame error between where the timer is drawn and where the pan really is	30m	Opus 5.5	high	P1
4.1	Baseline run: Chromium	Measure today's behaviour across clips, camera shapes and Gemini scenarios	20m	Sonnet 5.5	medium	P1
4.2	Baseline run: WebKit	Same, on the engine closest to Safari	15m	Sonnet 5.5	medium	P2
4.3	Triage	Match each measured failure to a code review finding; rank the fixes	15m	Opus 5.5	high	P1
5.1	Fix: camera crop mapping	Draw the timer in screen coordinates, not raw camera coordinates	30m	Opus 5.5	high	P1
5.2	Fix: hide when lost	Hide the timer when tracking fails instead of freezing it in place	20m	Opus 5.5	medium	P1
5.3	Fix: size ceiling	Stop the box growing frame on frame (the giant timer)	15m	Opus 5.5	medium	P1
5.4	Fix: re-seed blending	Stop each Gemini answer snapping the timer to a possibly wrong place	25m	Opus 5.5	high	P2
5.5	Fix: tracker aspect	Stop squashing the portrait camera feed into a 4:3 canvas	15m	Sonnet 5.5	medium	P2
5.6	Remove dead code	Delete what 1.2 found	10m	Sonnet 5.5	low	P3
6.1	Functional end-to-end	Everything that isn't AR: flows, page turn, swipe, timer controls, no-scroll	30m	Sonnet 5.5	medium	P2
7.1	Performance	CPU per frame under phone-level throttling, as a proxy for heat	15m	Sonnet 5.5	medium	P3
8.1	Report	Before and after numbers, what's left, what still needs your phone	10m	Sonnet 5.5	low	P1

Totals: P1 critical path ≈ 4h 05m. Everything ≈ 6h 10m.

If you're short on time, run the P1 rows in order and skip the rest. That path alone should answer whether the anchoring can be fixed.

Sub-phase details

0.1 Snapshot

- **Do:** create branch `testing-phase`, commit the current working tree. No push.
- **Needs you:** approve the commit.
- **Done when:** `git status` is clean; the commit hash is in the log.

1.1 Code review: AR pipeline

- **Scope:** `app/cook/page.tsx` (vision, anchor and render sections), `lib/useFlow.ts`, `lib/anchor.ts`, `lib/useVision.ts`, `app/api/look/route.ts`.
- **Confirm or rule out each known suspect:**
  1. Camera crop: boxes are in full-frame camera coordinates, but `object-cover` crops the feed to fit the screen, and the overlay is drawn as if screen = full frame.
  2. The tracker freezes at its last position when it loses the pan instead of hiding.
  3. Size compounds frame on frame with no absolute ceiling.
  4. Every Gemini answer re-seeds and snaps the timer, including inaccurate answers.
  5. The tracking canvas is a fixed 240x180 (4:3), squashing a portrait feed.
  6. The tilt is computed from a box whose shape is distorted by (1).
  7. Polling stops once Gemini confidently sees the pan (`settledRef`), so when the tracker loses the pan and freezes "waiting for the model", no re-detection ever comes. Together with (2) this is the likely cause of the timer stuck on a wall.
- **Out:** findings with `file:line`, severity and evidence, in the log. **No fixes.**

1.2 Dead code inventory

- **Scope:** `lib/usePan.ts`, `useStation.ts`, `useColour.ts`, `stationColour.ts`, `useMarkers.ts`, `markers.ts`, `useMotion.ts`, `app/markers/`.
- **Out:** each module marked used, unused or partly used, with a safe-to-delete verdict.

2.1 Install Playwright

- **Do:** `npm i -D @playwright/test`, then `npx playwright install chromium webkit` (a few hundred MB).
- **Done when:** a smoke test opens the home page in both browsers.

2.2 Synthetic test clips

- **Source:** `~/Downloads/IMG_9895.HEIC`, `IMG_9896.HEIC`, `IMG_9897.HEIC`. Convert to JPEG into `tests/fixtures/private/`.
- **Clips:** A static (control), B slow slide, C zoom in and out, D handheld shake, E pan leaves frame and returns, F no pan at all (a crop of wall or counter).
- **Each clip in three camera shapes:** 480x640 (4:3 portrait), 720x1280 (16:9 portrait), 640x480 (landscape). The shape matters because of suspect 1.
- **Formats:** `.y4m` for Chromium's fake camera, `.mp4` for the WebKit stub.
- **Ground truth:** a hand-checked pan box on each source photo, transformed per frame, saved as JSON in camera coordinates.
- **Limit:** this is 2D motion of still photos. It tests the mapping and tracker mechanics, not real 3D motion, steam or lighting. That's what your recording is for.

3.1 Playwright config

- iPhone 16 Pro: 402x874 viewport, scale factor 3, touch, mobile.
- Chromium flags: `--use-fake-ui-for-media-stream`, `--use-fake-device-for-media-stream`, `--use-file-for-fake-video-capture=<clip>`.
- Camera permission granted; video and trace recording on.
- **Done when:** the cook screen shows a clip as its camera.

3.2 Mock the vision route

- Intercept `/api/look` with `page.route`. Scenarios: accurate box (from ground truth), offset box (Gemini-style error), `panBox: null`, 503 then fallback, timeout, 15s delay.
- **Done when:** each scenario is selectable per test and reproducible.

3.3 Measurement

- Add `data-testid` to the ring and timer. Test-only attributes; no behaviour change.
- At each sample, read the drawn rectangle. Map the ground-truth box from camera to screen coordinates with **object-cover maths written independently in the test**, never imported from the app, so the measuring stick can't share the app's bug.
- **Per run:** centre error in px, size ratio, and visible-or-hidden. **Summary:** median and p90 error, % of frames on the pan, % of frames visible with no pan in shot.
- **Out:** a CSV per run, plus a summary.

4.1 Baseline run: Chromium

- Every clip x every camera shape x the key mock scenarios (accurate, offset, failure).
- **Out:** a summary table and run videos in `tests/results/baseline/`.

4.2 Baseline run: WebKit

- No fake-camera flags in WebKit: stub `getUserMedia` to return a canvas stream playing the `.mp4`.
- Smaller matrix: clips B, D and E, in two shapes.

4.3 Triage

- Map each baseline failure to a 1.1 finding: confirmed, refuted, or new.
- Re-rank the Phase 5 order if the numbers say otherwise.

Phase 5 fixes

Every fix follows the same loop: **implement, re-run the relevant baseline subset, record before and after numbers in the log, commit locally on your OK.**

- **5.1 Camera crop mapping:** convert camera-normalised boxes to screen coordinates using the object-cover scale and offset. Apply to the ring, the timer, and the tilt.
- **5.2 Hide when lost:** the tracker returns nothing after N failed frames; the timer and ring hide; no stale frozen box.
- **5.3 Size ceiling:** bound the size absolutely, relative to the seed box, not per frame.
- **5.4 Re-seed blending:** a new Gemini box eases in; one that disagrees badly with a confident track is rejected.
- **5.5 Tracker aspect:** size the tracking canvas to the camera's shape.
- **5.6 Remove dead code** found in 1.2.

6.1 Functional end-to-end

- Home, recipe, ingredients (omit onions and peppers), cook; steps reflect the omissions.
- Page turn forwards and backwards; swipe left and right; play and pause.
- Voice button present. Home and recipe don't scroll at 402x874; the ingredients back button stays fixed.

7.1 Performance

- Chromium with 4x CPU throttling. Measure tracker time per frame, achieved FPS, and long tasks. Flag anything over 33ms per frame.

8.1 Report

- Before and after metrics; what's fixed; what's left; a checklist for your phone.
- **Deploy is your decision**, made after reading this.

After your recording (separate, later)

ID	Sub-phase	What it's for	Est.	Model	Effort
9.1	Prepare real footage	Convert your clip; mark the pan in a sample of frames as ground truth	25m	Opus 5.5	high
9.2	Real-footage run	Run the same harness on real motion, steam and light; compare with the synthetic results	20m	Sonnet 5.5	medium